Bài 11 — Dataset Inspection

Số review có nhãn: 50000

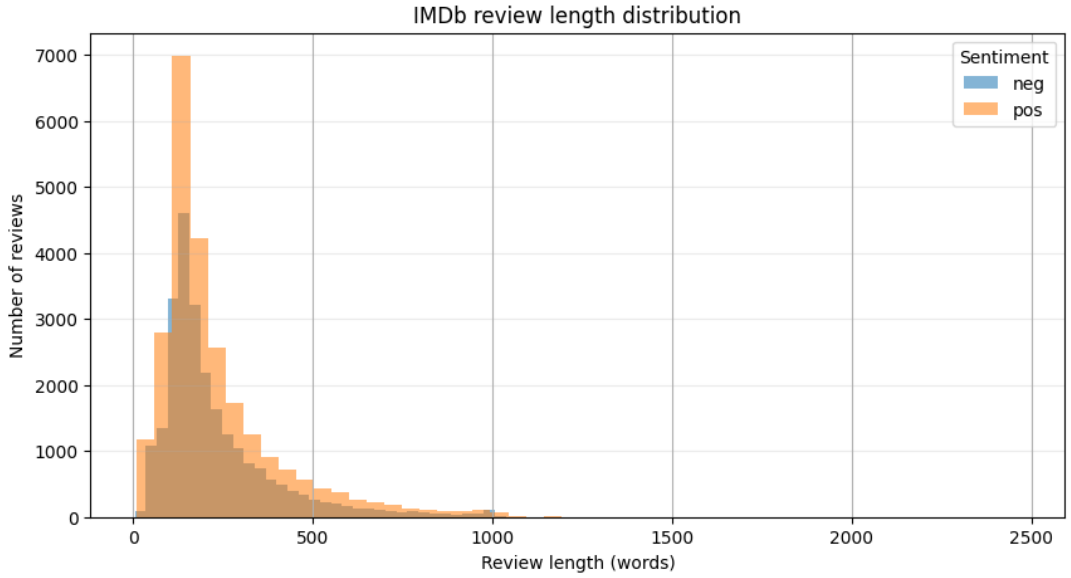
Số mẫu theo tập và nhãn:
label_name neg pos
split
test 12500 12500
train 12500 12500

Tỉ lệ nhãn toàn bộ dataset:
label_name
neg 0.5
pos 0.5
Name: proportion, dtype: float64

Số văn bản rỗng/thiếu: 0
Số nhãn thiếu: 0
Số dòng thuộc các review trùng nội dung: 824

Thống kê độ dài review (số từ):
count 50000.00
mean 231.16
std 171.34
min 4.00
25% 126.00
50% 173.00
75% 280.00
max 2470.00
Name: word_count, dtype: float64
Độ dài nhỏ nhất: 4
Độ dài trung vị: 173.0
Độ dài trung bình: 231.16
Độ dài lớn nhất: 2470

Biểu đồ phân bố độ dài review



Bài 12 — Tokenization và Padding

Kích thước vocabulary dùng cho model: 20000
Token PAD có ID: 0
Token OOV có ID: 1

Review mẫu (rút gọn): Story of a man who has unnatural feelings for a pig. Starts out with a opening scene that is a terrific example of absurd comedy. A formal orchestra audience is turned into an insane, violent mob by the crazy chantings of it's singers. Unfortunately it stays absurd the WHOLE time with no general nar ...
Token IDs trước padding (20 ID đầu): [63, 5, 4, 130, 35, 45, 7577, 1415, 16, 4, 4254, 515, 44, 17, 4, 634, 134, 13, 7, 4]
Sau padding/truncation về 20 vị trí: [63, 5, 4, 130, 35, 45, 7577, 1415, 16, 4, 4254, 515, 44, 17, 4, 634, 134, 13, 7, 4]

X_train shape: (25000, 200)
X_test shape: (25000, 200)
y_train shape: (25000,)
y_test shape: (25000,)
Giá trị padding: 0 | max sequence length: 200

Bài 13 — So sánh max_length

--- max_length = 50 ---
Epoch 1/3
313/313 - 18s - 58ms/step - accuracy: 0.7093 - loss: 0.5470 - val_accuracy: 0.7700 - val_loss: 0.4758
Epoch 2/3
313/313 - 22s - 71ms/step - accuracy: 0.8562 - loss: 0.3438 - val_accuracy: 0.7698 - val_loss: 0.5112
Epoch 3/3
313/313 - 17s - 53ms/step - accuracy: 0.9078 - loss: 0.2431 - val_accuracy: 0.7596 - val_loss: 0.6234

--- max_length = 100 ---
Epoch 1/3
313/313 - 32s - 102ms/step - accuracy: 0.7487 - loss: 0.4910 - val_accuracy: 0.7920 - val_loss: 0.4523
Epoch 2/3
313/313 - 26s - 85ms/step - accuracy: 0.8953 - loss: 0.2705 - val_accuracy: 0.8116 - val_loss: 0.5003
Epoch 3/3
313/313 - 28s - 89ms/step - accuracy: 0.9287 - loss: 0.1979 - val_accuracy: 0.7966 - val_loss: 0.7810

--- max_length = 200 ---
Epoch 1/3
313/313 - 53s - 170ms/step - accuracy: 0.7752 - loss: 0.4666 - val_accuracy: 0.8428 - val_loss: 0.3621
Epoch 2/3
313/313 - 81s - 260ms/step - accuracy: 0.9100 - loss: 0.2416 - val_accuracy: 0.8544 - val_loss: 0.4262
Epoch 3/3
313/313 - 51s - 163ms/step - accuracy: 0.9410 - loss: 0.1712 - val_accuracy: 0.8434 - val_loss: 0.5235

Bảng so sánh Bài 13:

max_length	tokens_retained_percent	training_time_seconds	accuracy	f1
50	20.9526	57.2766	0.7392	0.7430
100	40.5689	86.2652	0.7842	0.7986
200	67.4985	185.8351	0.8179	0.8119

Bài 14 — Cài đặt SimpleRNNCell

SimpleRNNCell output: [[0.05531239 0.16064717]]
PyTorch nn.RNN output: [[0.05531239 0.16064715]]
Sai khác tuyệt đối lớn nhất: 1.4901161193847656e-08
Đối chiếu thành công: kết quả custom cell khớp torch.nn.RNN.

Bài 15 — Baseline và Simple RNN

Train samples: 25000
Test samples: 25000
Train label counts (0=neg, 1=pos): [12500 12500]
Test label counts (0=neg, 1=pos): [12500 12500]

TF-IDF vectors unchanged after word reordering: True
Epoch 1/3
352/352 - 38s - 109ms/step - accuracy: 0.6654 - loss: 0.6056 - val_accuracy: 0.8204 - val_loss: 0.4349
Epoch 2/3
352/352 - 33s - 94ms/step - accuracy: 0.7395 - loss: 0.5213 - val_accuracy: 0.5236 - val_loss: 0.7816
Epoch 3/3
352/352 - 41s - 116ms/step - accuracy: 0.8253 - loss: 0.4066 - val_accuracy: 0.7164 - val_loss: 0.6816

Bảng kết quả Bài 15 (test):

Model	Training time (seconds)	Accuracy	Precision	Recall	F1
TF-IDF + Logistic Regression	0.6433	0.8836	0.8848	0.8821	0.8834
Embedding + Simple RNN	112.2563	0.7952	0.8510	0.7156	0.7775

RNN khi xáo trộn thứ tự token ở test (chỉ để phân tích):
Accuracy 0.7930
Precision 0.8447
Recall 0.7181
F1 0.7763

RNN accuracy change after shuffling: 0.0021

Bài 19 — So sánh RNN, LSTM và GRU

Vocabulary size: 20000
Train: (22500, 200) Validation: (2500, 200) Test: (25000, 200)
Label counts, train (neg=0, pos=1): [11250 11250]

========== RNN ==========
Epoch 1/3
352/352 - 39s - 110ms/step - accuracy: 0.5916 - loss: 0.6646 - val_accuracy: 0.5936 - val_loss: 0.6629
Epoch 2/3
352/352 - 37s - 104ms/step - accuracy: 0.7584 - loss: 0.5125 - val_accuracy: 0.7856 - val_loss: 0.4868
Epoch 3/3
352/352 - 37s - 105ms/step - accuracy: 0.8198 - loss: 0.4110 - val_accuracy: 0.8000 - val_loss: 0.4699

========== LSTM ==========
Epoch 1/3
352/352 - 70s - 199ms/step - accuracy: 0.7613 - loss: 0.4883 - val_accuracy: 0.8460 - val_loss: 0.3604
Epoch 2/3

352/352 - 68s - 192ms/step - accuracy: 0.8972 - loss: 0.2756 - val_accuracy: 0.8508 - val_loss: 0.3816
Epoch 3/3
352/352 - 68s - 192ms/step - accuracy: 0.9271 - loss: 0.2038 - val_accuracy: 0.8212 - val_loss: 0.4922

========== GRU ==========
Epoch 1/3
352/352 - 74s - 210ms/step - accuracy: 0.7673 - loss: 0.4690 - val_accuracy: 0.8328 - val_loss: 0.4061
Epoch 2/3
352/352 - 81s - 231ms/step - accuracy: 0.8950 - loss: 0.2769 - val_accuracy: 0.8380 - val_loss: 0.3953
Epoch 3/3
352/352 - 83s - 237ms/step - accuracy: 0.9291 - loss: 0.1973 - val_accuracy: 0.8200 - val_loss: 0.4982

Bảng so sánh Bài 19 (test set):

Model	Parameters	Training time (seconds)	Accuracy	Precision	Recall	F1
RNN	1288321	116.5525	0.7946	0.8078	0.7732	0.7901
LSTM	1313089	205.5947	0.8109	0.7559	0.9184	0.8293
GRU	1305025	238.5716	0.8107	0.7480	0.9371	0.8320

Bài 20 — Ảnh hưởng của sequence length

Vocabulary size: 20000
Train/validation/test samples: 22500 2500 25000
Tổng token trong train trước truncation: 5342796

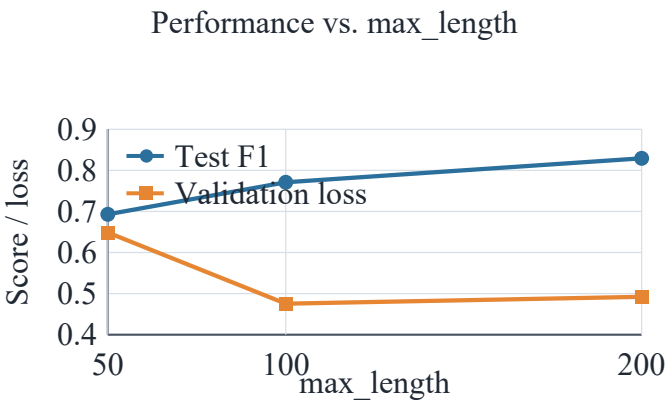
========== max_length = 50 ==========
Epoch 1/3
352/352 - 23s - 65ms/step - accuracy: 0.7260 - loss: 0.5290 - val_accuracy: 0.7764 - val_loss: 0.4748
Epoch 2/3
352/352 - 20s - 57ms/step - accuracy: 0.8542 - loss: 0.3456 - val_accuracy: 0.7780 - val_loss: 0.5421
Epoch 3/3
352/352 - 19s - 53ms/step - accuracy: 0.9020 - loss: 0.2532 - val_accuracy: 0.7564 - val_loss: 0.6487

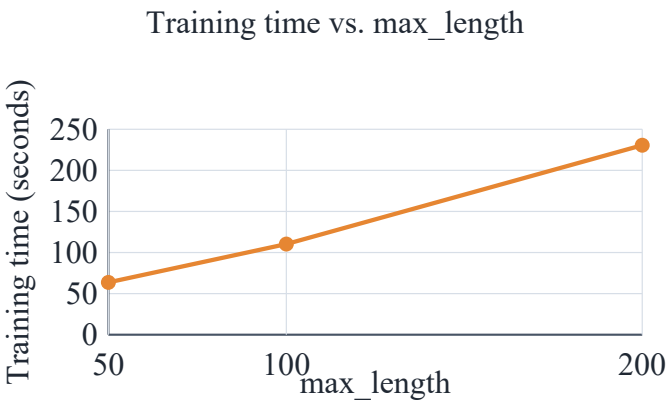
========== max_length = 100 ==========
Epoch 1/3
352/352 - 36s - 102ms/step - accuracy: 0.7603 - loss: 0.4815 - val_accuracy: 0.8092 - val_loss: 0.4499
Epoch 2/3
352/352 - 33s - 95ms/step - accuracy: 0.8892 - loss: 0.2847 - val_accuracy: 0.7920 - val_loss: 0.5666
Epoch 3/3
352/352 - 34s - 96ms/step - accuracy: 0.9120 - loss: 0.2347 - val_accuracy: 0.7932 - val_loss: 0.4755

========== max_length = 200 ==========
Epoch 1/3
352/352 - 66s - 186ms/step - accuracy: 0.7613 - loss: 0.4883 - val_accuracy: 0.8460 - val_loss: 0.3604
Epoch 2/3
352/352 - 81s - 230ms/step - accuracy: 0.8972 - loss: 0.2756 - val_accuracy: 0.8508 - val_loss: 0.3816
Epoch 3/3
352/352 - 84s - 239ms/step - accuracy: 0.9271 - loss: 0.2038 - val_accuracy: 0.8212 - val_loss: 0.4922

Kết quả Bài 20:

max_length	tokens_retained_percent	training_time_seconds	validation_loss	test_f1	padded_input_memory_mb
50	20.9754	63.6511	0.6487	0.6930	9.5367
100	40.5999	110.3022	0.4755	0.7710	19.0735
200	67.5904	230.5821	0.4922	0.8293	38.1470





Bài 21 — Long-range dependency

Vocabulary size: 20000
Train/validation/test: (22500, 200) (2500, 200) (25000, 200)

Epoch 1/3
352/352 - 38s - 108ms/step - accuracy: 0.5916 - loss: 0.6646 - val_accuracy: 0.5936 - val_loss: 0.6629
Epoch 2/3
352/352 - 33s - 93ms/step - accuracy: 0.7584 - loss: 0.5125 - val_accuracy: 0.7856 - val_loss: 0.4868
Epoch 3/3
352/352 - 34s - 98ms/step - accuracy: 0.8198 - loss: 0.4110 - val_accuracy: 0.8000 - val_loss: 0.4699

Test accuracy: 0.7946
Test F1: 0.7901
Training time (seconds): 111.81

Các câu kiểm soát có cùng độ dài:

true_sentiment	position	sequence_tokens	p_positive	predicted_sentiment	correct
positive	beginning	194	0.9392	positive	True
positive	middle	194	0.9458	positive	True
positive	end	194	0.9465	positive	True
negative	beginning	194	0.9383	positive	False
negative	middle	194	0.9460	positive	False
negative	end	194	0.7350	positive	False

Trung bình xác suất positive theo vị trí:

position	negative	positive
beginning	0.9383	0.9392
middle	0.9460	0.9458
end	0.7350	0.9465

Bài 23 — Learning Curves

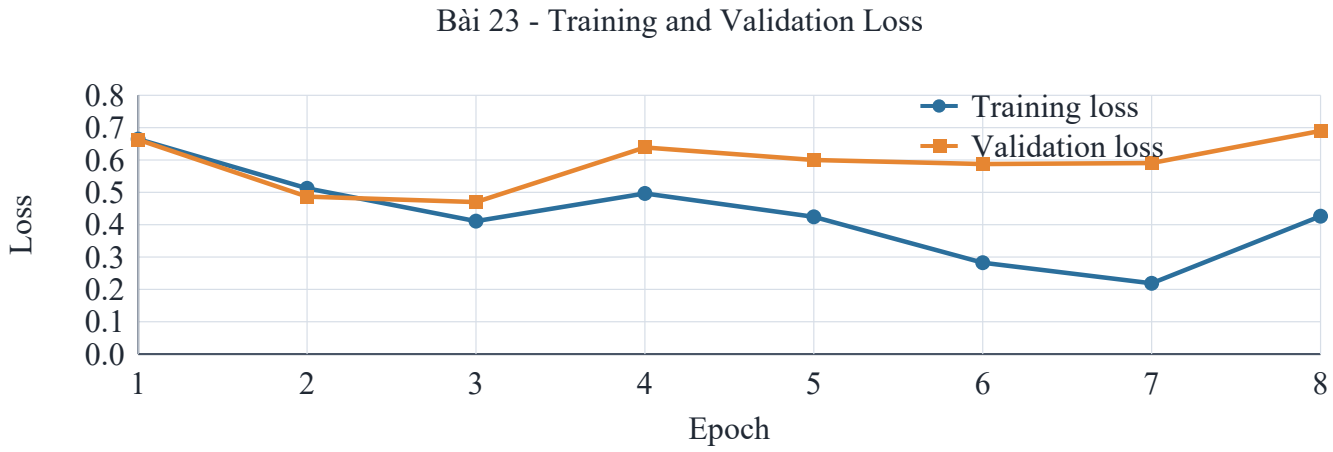
Vocabulary size: 20000
Train shape: (22500, 200) Validation shape: (2500, 200)

Epoch 1/8
352/352 - 40s - 113ms/step - accuracy: 0.5916 - loss: 0.6646 - val_accuracy: 0.5936 - val_loss: 0.6629
Epoch 2/8
352/352 - 37s - 105ms/step - accuracy: 0.7584 - loss: 0.5125 - val_accuracy: 0.7856 - val_loss: 0.4868
Epoch 3/8
352/352 - 41s - 116ms/step - accuracy: 0.8198 - loss: 0.4110 - val_accuracy: 0.8000 - val_loss: 0.4699
Epoch 4/8
352/352 - 40s - 115ms/step - accuracy: 0.7493 - loss: 0.4966 - val_accuracy: 0.6316 - val_loss: 0.6388
Epoch 5/8
352/352 - 40s - 114ms/step - accuracy: 0.8012 - loss: 0.4243 - val_accuracy: 0.7856 - val_loss: 0.6000
Epoch 6/8
352/352 - 38s - 107ms/step - accuracy: 0.8890 - loss: 0.2825 - val_accuracy: 0.7988 - val_loss: 0.5875
Epoch 7/8
352/352 - 38s - 107ms/step - accuracy: 0.9197 - loss: 0.2188 - val_accuracy: 0.8128 - val_loss: 0.5906
Epoch 8/8
352/352 - 41s - 115ms/step - accuracy: 0.8070 - loss: 0.4263 - val_accuracy: 0.6436 - val_loss: 0.6900

Learning curves theo epoch:

Epoch	Train Loss	Validation Loss
1	0.6646	0.6629
2	0.5125	0.4868
3	0.4110	0.4699
4	0.4966	0.6388
5	0.4243	0.6000
6	0.2825	0.5875
7	0.2188	0.5906
8	0.4263	0.6900

Validation loss thấp nhất ở epoch: 3
Train loss epoch cuối: 0.4263
Validation loss epoch cuối: 0.69
Đánh giá overfitting bằng cách xem train loss có giảm trong khi validation loss tăng sau epoch tốt nhất hay không.



Bài 26 — Sentiment Classifier Application

Mô hình Embedding → LSTM → Dense được huấn luyện trên tập IMDb; đầu vào được token hóa, padding/truncation về 200 token.

Kiến trúc mô hình

Layer	Output shape	Tham số
Embedding	(None, 200, 64)	1,280,000
LSTM	(None, 64)	33,024
Dense	(None, 1)	65
Tổng số tham số		1,313,089

Kết quả huấn luyện

Epoch	Train accuracy	Train loss	Val. accuracy	Val. loss
1	0.7555	0.5006	0.8316	0.4183
2	0.8973	0.2636	0.8444	0.3803
3	0.9270	0.2037	0.7600	0.5090

Validation loss thấp nhất ở epoch 2 (0.3803), với validation accuracy 0.8444. Ở epoch 3, train loss tiếp tục giảm nhưng validation loss tăng lên 0.5090 và validation accuracy giảm còn 0.7600.

Dự đoán các review mẫu

Loại review	Nhãn kỳ vọng	P(Positive)	Dự đoán	Đúng
Ngắn	Positive	0.7280	Positive	Có
Dài	Positive	0.9620	Positive	Có
Phủ định	Negative	0.1331	Negative	Có
Tương phản	Positive	0.9171	Positive	Có

Cả bốn review mẫu được phân loại đúng theo nhãn kỳ vọng; review phủ định có xác suất tích cực thấp nhất (0.1331).